MIL-C-83396(USAF)
AMENDMENT 2
7 FEBRUARY 1986
SUPERSEDING
AMENDMENT 1
3 MARCH 1978

MILITARY SPECIFICATION
CONTROL ASSEMBLY, GYROSCOPE, ATTITUDE
HEADING A/A24G-42

This amendment forms a part of MIL-C-83396(USAF), dated 30 March 1973, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

* Add:
"1.1 title "Scope."

* Delete "2.1" and add: "2.1 Government documents".

* Add:
"2.1.1 Specifications standards, and handbooks. The following specifications, standard, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATIONS Federal add:
"PPP-B-601 Boxes, Wood, Cleated Plywood" and "PPP-B-636 Boxes , Fiberboard".

* Under SPECIFICATIONS Military in title of MIL-C-5015 delete "AN Type" and add "Circular Threaded, An Type, General Specification for " and delete "MIL-C-25773".

Delete "aircraft" and add "aerospace" in title of MIL-E-5400.

* Add: "MIL-M-2793/6 , Meter, Time Totalizing Minature, Digital, 115 Volts, 400 Hz".

* Under STANDARDS Military delete "MIL-STD-100" and add "DOD-STD-100".

PAGE 2

* Under STANDARDS Military in MIL-STD-704 delete "title" and add "Aircraft, Electric Power Characteristics" and MIL-STD-810 title add "and Engineering Guidelines" at the end of sentence. Between "Electric" and "Solder" and "Series 1,". Delete "MS17322".

* Add boxed paragraph to bottom center of page:

"Beneficial comments (recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

FSC 6610

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Standardization Document Improvement Proposal (DD Form 1426)
Check the source to verify that this is the current version before use.

* Delete source paragraph and add:
(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting activity.)

* Delete 2.2 and add:
"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation."

Immediately below North Specification add:
"Exhibits

ASD/ENA1-76-26 Indicator, Horzontal Situation AQU- 13A/A".

* In source state of Non-Government delete "procuring" and add "contracting".

PAGE 3

3.2.1.2, add:

"e. AQU-13A/A Horizontal Situation Indicator (Exhibit ASD/ENA 1-76-26)

(Application for copies should be addressed to ASD/ENA 1, Wright-Patterson AFB, OH 45433 or to the contracting activity.)"

PAGE 4

Table I, Pins F and G: Change to "SPARE".

Pin H: Change to "Pitch and Roll Election".

Pin K: Change to "SPARE".

Pin R: Change to "SPARE".

Pin U: Change to "SPARE".

PAGE 5

Table II, Pins N, P, R and T: Redesignate "SPARE".

PAGE 6

* Add:
"3.4.3.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

PAGE 8

"3.5.3.1.4, Line 8: Delete "be brought out of the case so that precession

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

voltage can be applied externally to precess the gyro for plane rate and earth rate correction." and substitute " be connected so that internal pitch and roll acceleration sensors shall control the erection of the gyro."

3.5.3.1.4.3, Line 5: Delete "the gyro in roll $\pm 45^{o}$ or less". and substitute "the gyro case in roll $\pm 45^{o}$ or less".

3.5.3.1.4.4, Add after last sentence: Fast erection shall override erection cut-out in either pitch or roll.

PAGE 10

3.5.3.1.4.5: Delete and substitute:
"3.5.3.1.4.5 Pitch and roll erection cut-out. The design shall be such that when Pins H and S of Table I are connected externally, both pitch bank erection shall be functioning. Pitch and roll erection shall be cut-out when gimbal mounted acceleration sensors detect an acceleration equal to or greater than 0.1g ±15% in fore-aft and lateral aces respectively. Pitch and roll erection cut-out shall be independent of each other and shall be overridden when fast erection is applied."

3.5.3.1.4.6, Line 1: Delete "For test purposes, a means shall be provided" and substitute "Means shall be provided".

PAGE 15

3.5.3.2.1.1, Line 6: Delete "the gimbal shall have a minimum of $\pm 60^{o}$" and substitute "The gimbal shall have a minimum of $\pm 30^{o}$".

PAGE 16

Add:
"3.5.3.2.3.1 DG slaving and leveling gimbal cut-out. Provisions shall be made to cut out DG slavings and DG inner gimbal leveling when either pitch erection and roll erection cut-out take place. Leveling cut-out shall operate under all modes of operation. Slaving cut-out shall be effective when the system is operating in the magnetic mode."

3.5.3.2.4.2 Delete in its entirety.

PAGE 17

3.5.3.2.4.3.2 Delete in its entirety.

PAGE 20

* 3.5.3.9 Delete "MS17322-8" and add "MIL-M-7793/6".

PAGE 28

* 3.5.5.1 Delete "conforming to MIL-C-25773" and add "period" at the end of sentence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

PAGE 37

4.6.1.1 Delete and substitute:
"4.6.1.1 Examination of product. Prior to sealing, the displacement gyroscope shall be inspected to determine compliance with the requirements of 3.4.3.12, 3.13, and 3.15. Upon sealing, the gyroscope shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, markings, weight, and outline dimensions. The case shall be inspected for proper painting and general good workmanship, and the connector shall be inspected to make certain that the pins are not bent or damaged. The above data shall be recorded. The weight and dimensions requirements need only be recorded once for each gyro produced."

PAGE 38

4.6.1.5 Line 1 Delete "between 45° and".

Line 4 Delete "5° on" and substitute "from 5° ti 1° in".

4.6.1.6: Delete and substitute:
"4.6.1.6 Erection and slaving cut-out. Pitch and roll erection cut-off shall be tested as follows:

a. With the system properly connected, mount the displacement gyroscope on a table level to within 0.05° and point it to a North heading. Apply power and allow the system to stabilize.

b. Precess the vertical gyroscope to a pitch angel of 4.8°. Pitch, roll erection and slaving shall be ON.

c. Precess the vertical gyroscope to a pitch angle of 6.6°. Pitch erection and slaving shall be cut-off. Roll erection shall remain ON.

d. Repeat steps b and c except that the pitch precession shall be in the negative direction.

e. With the gyro still pointing North and the pitch precessed to -6.6° rotate the gyro so that it is pointed West. Roll erection and slaving shall be cut-off. Pitch erection shall remain ON.

f. Precess the vertical gyroscope to a roll angle of -4.8° in the same manner as described in e. Pitch and roll erection and slaving shall be ON.

g. Repeat steps e and f except that the roll precession shall be in a positive direction."

PAGE 40

4.6.1.15, Line 7: Delete "4 hours" and substitute "1 hour".

Line 9: Delete last sentence.

4.6.1.16, Line 1: Delete "Latitude correction" and substitute "Latitude corrected DG drift".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

Line 3: Delete "0°" and substitute "local latitude".

Line 4: Delete "4.6.1.15. The drift of the gyro for each hourly period shall not differ by" and substitute "4.6.1.15. The DG drift for each half hour period shall not differ by".

4.6.1.17: Delete in its entirety.

PAGE 41

4.6.1.20.2.a, Second sentence: Delete "servoed".

Third sentence: Delete "servoed".

4.7.1.20.2.d, Line 3: Delete "to the percentage values on the voltage ratio bridges for sine and cosine outputs listed in the third and fourth columns from the left, respectively." and substitute "to the percentage values on the voltage ratio bridges for sine or cosine outputs listed.".

Line 6: Delete "servoed".

PAGE 43

Table VI: Delete and substitute table VI as shown on page 5 herein.

PAGE 44

4.6.1.22, Column headed "From connector pins":

Line 1: Delete "J 8/N".

Line 2: Delete "F".

Line 6: Delete.

Line 7: Delete.

Column headed "To connector pins of case":

Line 6: Delete.

Line 7: Delete.

PAGE 45

4.6.2.1: Delete and substitute:
"4.6.2.1 Examination of product. Prior to final assembly, the amplifier shall be examined to determined compliance with the requirements of 3.4, 3.12, 3.13, and 3.15. After assembly, the amplifier shall be examined for compliance with the requirements specified herein with respect to materials, workmanship, marking, weight and dimensions. The case shall be inspected for proper painting and general good workmanship and the connector shall be inspected to make sure that the pins are not bent or damaged. The above data shall be recorded. The weight and dimensions requirements need only be recorded once for each amplifier produced."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

TABLE VI. Percent voltage ratio for cw roll

ANGLE	NOMINAL INPUT SIN	NOMINAL INPUT COS	READOUT - API SIN 1	READOUT - API COS 1	READOUT - API SIN 2	READOUT - API COS 2	READOUT - API SIN 3	READOUT - API COS 3	READOUT - API SIN 4	READOUT - API COS 4	API LIMITS INDICATOR LOW	API LIMITS INDICATOR HIGH
0	0000		—		—		—		—		-0°43'	0°43'
5	0.872		—		—		—		—		4°17'	5°43'
15	2.588		—		—		—		—		14°17'	15°43'
30	5.000		—		—		—		—		29°17'	30°47'
60		5.000		—		—		—		—	59°13'	60°47'
90		0.000		—		—		—		—	89°17'	90°43'
120		5.000		—		—		—		—	119°13'	120°47'
135		7.071		—		—		—		—	134°6'	135°54'
180	0.000		—		—		—		—		179°17'	-179°17'
-135		7.071		—		—		—		—	-135°54'	-134°6'
-120		5.000		—		—		—		—	-120°47'	-119°13'
-90		0.000		—		—		—		—	-90°43'	-89°17'
-60		5.000		—		—		—		—	-60°47'	-59°13'
-30	5.000		—		—		—		—		-30°47'	-29°13'
-15	2.588		—		—		—		—		-15°43'	-14°17'
-5	0.872		—		—		—		—		-5°43'	-4°17'
0	0.000		—		—		—		—		-0°43'	-0°43'

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

PAGE 57

4.6.11.1, Line 3: Delete "504" and substitute "504.1".

Line 3: Delete "class 2 except the pressure in the chamber shall be the equivalent of 80,000 ±500 feet for steps 3, 11, and 12." and substitute "Category 6".

4.6.11.2, Line 3: Delete "501" and substitute "501.1".

4.6.11.3, Line 3: Delete "502" and substitute "502.1".

4.6.11.4, Line 2: Delete "503" and substitute "503.1".

PAGE 58

4.6.11.5.2, First sentence: Delete "method 514, equipment category (b), procedure I (with vibration isolators, part 1 only), except the curve shown on figure 8 herein shall apply." and substitute "514.2 category b.2".

4.6.11.7 Line 5: Delete "516.1-2" and substitute "516.2-2".

4.6.11.8 Line 2: Delete "507" and substitue "507.1".

PAGE 61

FIGURE 8: Delete in its entirety.

PAGE 62

4.6.11.9 Line 2: Delete "508" and substitute "508.1".

4.6.11.10 Line 2: Delete "506" and substitute "506.1".

4.6.11.11 Line 2: Delete "509" and substitute "509.1".

4.6.11.12 Line 2: Delete "510" and substitute "510.1".

4.6.11.13 Line 2: Delete "511" and substitute "511.1".

PAGE 65

* Delete "4.7" and add:
"4.7 Inspection and Test. Tests of Methods of preservation and packing shall be accomplished in accordance with Section 4 MIL-P-116 to insure compliance with Section 5 of this specification."

* Delete Section 5 and add:
"5. PACKING

5.1 Preservation-Packaging. Preservation-packaging shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

5.1.1 Cleaning. Units shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Units shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each control assembly shall be packaged in quantity unit pack of one each. Nonhermetically sealed components shall be preserved in accordance with MIL-P-116, Method II, and hermetically sealed components shall be preserved in accordance with Method III. Electrical connectors on hermetically sealed components shall be provided with water-vaporproof caps and gaskets. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in Gs established for the control assemblies when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Units shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of control assemblies shall be in accordance with ASTM-D-3951-82.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment form the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged control assemblies shall be packed in accordance with ASTM-D-3951-82.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.

5.4 Reinspection Date. The reinspection date markings shall be as specified by the contracting activity."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodian:
Air Force - 99

Preparing Activity
Air Force - 71

Project No.
6610-F058

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:54Z
Check the source to verify that this is the current version before use.